Single Element in a sorted array.

i/p arr : [1, 1, 2, 3, 3, 4, 4, 8, 8]

(e, o) (e) (o, e) (o, e)

output = 2

ans always at even index

Ans
- left (e, o)
- right (o, e)

Logic 2 cases even / odd

```
if ( start == end ) return arr[mid];
```

even

(e, o) → if (arr[mid] == arr[mid+1])
{ start = mid + 2; } — 2 because we already check mid+1 ele.

else if (mid == 0) → case :
{ return arr[mid]; }

(o, e) else {
if (arr[mid] != arr[mid-1]) ⇒ at above condⁿ ele ≠ mid+1 & in this ele ≠ mid-1, that means is is ans.
{ return arr[mid]; }
else { end = mid; } ⇒ end = mid because
[1, 1, 2, 2, 3, 3, 4, 5, 5, 6, 6]
if we at ans so end should also be at mid to satisfy (start == end) (we avoid to lost track)
}

odd

[1, 1, 2, 2, 3, 3, 3 . . .]
0 1 2 3 4 5

Odd

(e, o) → if (arr[mid] == arr[mid-1])
{ start = mid + 1; } because we only check mid-1 index.

(o, e) else { end = mid }

end